

Condition (A): log-concavity of cylindric Kostka numbers at $\ell = 3$

A reformulation valid for all m , a complete reduction at $m = 2$,

and a proof in the single-region case

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Abstract

Let $s_{\lambda/\mu}^c(x_1, x_2, x_3)$ be a cylindric skew Schur polynomial at level n with m beads, and let $k(a, b) = K_{(a, b, d-a-b)}^c$ be its cylindric Kostka numbers. Condition (A) is the assertion $k(a, b)^2 \geq k(a+1, b)k(a-1, b)$; by the S_3 -symmetry of s^c this is log-concavity along every root direction, and it is one of the two conjuncts that would settle the Lorentzian property (L3) at $\ell = 3$.

We prove three things and record one gap. (i) For *every* m , condition (A) is equivalent to the statement that a sum of PF_2 sequences over a single slice of a box is log-concave (Proposition 2.1); the summands are exactly the $\ell = 2$ cylindric Kostka sequences, which are PF_2 by an already-proved theorem. This converts (A) from a statement about a three-step chain into a one-parameter statement. (ii) At $m = 2$ the slice is one-dimensional, and (A) reduces *exactly* to a self-contained combinatorial lemma about a concave and a convex 1-Lipschitz integer profile (Proposition 3.1, Conjecture 3.2). The 1-Lipschitz hypothesis is sharp: dropping it produces counterexamples. (iii) At $m = 2$ the parameter range splits at two explicit breakpoints into at most four regions, each of which contributes a PF_2 sequence, by two different mechanisms which we prove (Theorem 3.6). Consequently (A) holds whenever a single region is active — an explicitly checkable criterion satisfied by 79.9% of the slices in our sweep (Corollary 3.7).

The remaining gap is precisely stated: the pairwise mixed inequality between two *distinct* regions (verified on 525 of 525 region pairs, unproved).

We emphasise what is *not* claimed. Condition (A) alone does not settle (L3), even at $\ell = 3$: by **13-det-reduction**, (L3) at $\ell = 3$ is equivalent to $e_2(M) \leq 0$ and $\det M \geq 0$, and (A) delivers only the first conjunct.

1 Setting and statement

1.1 Cylindric shapes

Fix a level $n \geq 2$ and a number of beads $m \geq 1$. A *cylindric shape* is a tuple $x = (x_1, \dots, x_m) \in \mathbb{Z}^m$ with $x_1 < x_2 < \dots < x_m < x_1 + n$, extended cyclically by $x_{i+m} = x_i + n$. For cylindric shapes ν, ρ we write $\nu \prec \rho$ (ρ/ν is a *horizontal strip*) if

$$\nu_i \leq \rho_i \leq \nu_{i+1} - 1 \quad (1 \leq i \leq m),$$

cyclically, and set $|\rho/\nu| = \sum_i (\rho_i - \nu_i)$. For a cylindric skew shape λ/μ (so $\mu_i \leq \lambda_i$ for all i) with $d = |\lambda/\mu|$, the cylindric Kostka number $K_{(a_1, \dots, a_\ell)}^c$ counts chains $\mu = \rho^0 \prec \rho^1 \prec \dots \prec \rho^\ell = \lambda$ with $|\rho^t/\rho^{t-1}| = a_t$, and $s_{\lambda/\mu}^c(x_1, \dots, x_\ell) = \sum_\alpha K_\alpha^c x^\alpha$.

Throughout, $\ell = 3$ and $k(a, b) := K_{(a, b, d-a-b)}^c$. Since s^c is symmetric, $k(a, b) = k(b, a)$.

Conjecture 1.1 (Condition (A)). *For every cylindric skew shape λ/μ and all a, b , $k(a, b)^2 \geq k(a+1, b)k(a-1, b)$.*

1.2 Inputs used

We use the following, all recorded as `proved` in the registry `proofs/registry/cylindric-lorentzian.json`.

- (P1) (`interlacing-model`) At $\ell = 3$, $K_{(a, b, c)}^c$ counts pairs $(\kappa, \sigma) \in \mathbb{Z}^m \times \mathbb{Z}^m$ with $\mu_i \leq \kappa_i \leq \mu_{i+1} - 1$, $\lambda_{i-1} + 1 \leq \sigma_i \leq \lambda_i$ and $\kappa_i \leq \sigma_i \leq \kappa_{i+1} - 1$ (cyclically), graded by $a = S_\kappa - |\mu|$, $b = S_\sigma - S_\kappa$, $c = |\lambda| - S_\sigma$.
- (P2) (`ell2-decoupling`) At $\ell = 2$ the constraints on the single intermediate shape decouple into one independent interval per bead.
- (P3) (`pf2-convolution`) PF_2 — log-concave with interval support — is closed under convolution.
- (P4) (`ell2-all-m`) $\sum_a K_{\lambda/\mu, (a, d-a)}^c z^a$ has PF_2 coefficients, for every cylindric skew shape, every n , every m .

We write PF_2 for: nonnegative, support an interval of $\mathbb{Z}$, and $g(s)^2 \geq g(s-1)g(s+1)$ for all s .

Lemma 1.2 (concave $\Rightarrow \text{PF}_2$). *Let $g : \mathbb{Z} \rightarrow \mathbb{Z}_{\geq 0}$ have support an interval $J = [j_0, j_1]$ and satisfy $2g(s) \geq g(s-1) + g(s+1)$ for all $j_0 < s < j_1$. Then $g \in \text{PF}_2$.*

Proof. For $j_0 < s < j_1$, AM–GM gives $g(s) \geq \frac{1}{2}(g(s-1) + g(s+1)) \geq \sqrt{g(s-1)g(s+1)}$, so $g(s)^2 \geq g(s-1)g(s+1)$. For $s = j_0$ we have $g(j_0-1) = 0$, so the right side vanishes; likewise at $s = j_1$. For $s \notin J$ at least one of $g(s-1), g(s+1)$ lies outside J or $g(s)$ is on the boundary; in every case $g(s-1)g(s+1) = 0$ unless $s-1, s+1 \in J$, which forces $s \in J$. The support is an interval by hypothesis. $\square$

Lemma 1.3 (truncated concave). *Let $c : \mathbb{Z} \rightarrow \mathbb{Z}$ be concave (that is, $2c(s) \geq c(s-1) + c(s+1)$ for all s). Then $g = \max(c, 0)$ has interval support and is PF_2 .*

Proof. Put $J = \{c > 0\}$.

J is an interval. A concave integer sequence is quasiconcave: $c(s) \geq \min(c(r), c(t))$ whenever $r \leq s \leq t$. (Induction on the span $t-r$: for $r < s < t$ apply the statement to $[r, t-1]$ and to $[r+1, t]$ and compare the two slopes, using that increments $c(\cdot+1) - c(\cdot)$ are non-increasing.) So $c(r) > 0$ and $c(t) > 0$ force $c(s) > 0$ for every s between them, which is exactly interval support.

Log-concavity. Fix s . If $c(s-1) \leq 0$ then $g(s-1) = 0$ and $g(s-1)g(s+1) = 0 \leq g(s)^2$; likewise if $c(s+1) \leq 0$. If $c(s-1) > 0$ and $c(s+1) > 0$ then quasiconcavity gives $c(s) > 0$ as well, so g agrees with c at all three points, and $2c(s) \geq c(s-1) + c(s+1)$ with $c(s \pm 1) \geq 0$ gives $c(s)^2 \geq \frac{1}{4}(c(s-1) + c(s+1))^2 \geq c(s-1)c(s+1)$ by AM–GM.

Note that this argument does *not* pass through Lemma 1.2. That lemma requires the support to be a *bounded* interval $J = [j_0, j_1]$, and $\max(c, 0)$ need not have bounded support: $c \equiv 1$ is concave with $\{c > 0\} = \mathbb{Z}$, so there is no j_1 . The proof above needs no boundedness. $\square$

Remark 1.4. Lemma 1.3 is machine-checked, for $c : \mathbb{Z} \rightarrow \mathbb{Z}$, as `TworowD4Kernel.DiscreteConcavity.PFtwo_po` in `lean/tworow_d4_kernel/TwoRowD4Kernel/DiscreteConcavity.lean` (axioms: `propext`, `Quot.sound` only); the quasiconcavity step is `IntConcave.min_le` and the counterexample to citing Lemma 1.2 is `posPart_support_unbounded`. The earlier version of this proof read “at the two ends of the support the argument of Lemma 1.2 applies verbatim”, and also allowed c to take the value $-\infty$ off an interval. The first clause was a citation whose object is undefined in the unbounded case; the second is not needed, since at the only place Lemma 1.3 is used — w_I and w_{III} in the proof of Proposition 3.4 — the function c is a genuine $\mathbb{Z}$ -valued concave sequence, being minus a maximum of finitely many affine functions.

2 A reformulation of (A), valid for every m

Proposition 2.1. Let $\text{Box}(\mu) = \prod_{i=1}^m [\mu_i, \mu_{i+1} - 1] \subset \mathbb{Z}^m$ (with $\mu_{m+1} = \mu_1 + n$), and for $\nu \in \text{Box}(\mu)$ put

$$L_i(\nu) = \max(\nu_i, \lambda_{i-1} + 1), \quad R_i(\nu) = \min(\nu_{i+1} - 1, \lambda_i), \quad f_\nu = \bigstar_{i=1}^m \mathbf{1}_{[L_i(\nu) - \nu_i, R_i(\nu) - \nu_i]}.$$

Then for all a, b ,

$$k(a, b) = k(b, a) = \sum_{\substack{\nu \in \text{Box}(\mu) \\ S_\nu = |\mu| + b}} f_\nu(a),$$

and each f_ν is PF₂. Consequently condition (A) is equivalent to: for every b , the sum of the PF₂ sequences f_ν over the box-slice $\{\nu \in \text{Box}(\mu) : S_\nu = |\mu| + b\}$ is log-concave.

Proof. By (P1) with the roles of the two interior layers named ν, κ , $k(b, a)$ counts pairs with $|\nu/\mu| = b$, $|\kappa/\nu| = a$. The condition $\mu \prec \nu$ reads $\mu_i \leq \nu_i \leq \mu_{i+1} - 1$, one independent interval per bead, so the admissible ν form exactly the box $\text{Box}(\mu)$; no separate shape check is needed, because $\nu_i \leq \mu_{i+1} - 1 < \mu_{i+1} \leq \nu_{i+1}$. Given ν , the conditions $\nu \prec \kappa$ and $\kappa \prec \lambda$ read $\nu_i \leq \kappa_i \leq \nu_{i+1} - 1$ and $\lambda_{i-1} + 1 \leq \kappa_i \leq \lambda_i$, i.e. $\kappa_i \in [L_i(\nu), R_i(\nu)]$, again one independent interval per bead; and κ is automatically a cylindric shape because $\kappa_i \leq \nu_{i+1} - 1 < \nu_{i+1} \leq \kappa_{i+1}$. Grading by $a = \sum_i (\kappa_i - \nu_i)$ gives the convolution f_ν . Each f_ν is a convolution of interval indicators, hence PF₂ by (P3); this is (P2)+(P4) for the shape λ/ν . Finally $k(a, b) = k(b, a)$ by the symmetry of s^c . $\square$

Remark 2.2. The point of Proposition 2.1 is that it removes one of the two interior layers. Condition (A) is *not* a statement about a three-step chain; it is the statement that a particular finite family of PF₂ sequences, indexed by the lattice points of a box lying on one hyperplane, sums to a log-concave sequence.

2.1 What this rules out

The following were tested against Proposition 2.1 and *fail*; each is recorded with the scope of its reason, since each is a route that looks open until it is tried.

1. **Pairwise comparison inside a slice is not enough.** For sequences $g_1, \dots, g_N \geq 0$, expanding $(\sum g)^2 - (\sum g)(s-1)(\sum g)(s+1)$ and symmetrising shows that $\sum g_j$ is log-concave as soon as each g_j is log-concave and every pair satisfies

$$2g_j(s)g_k(s) \geq g_j(s-1)g_k(s+1) + g_j(s+1)g_k(s-1) \quad \text{for all } s. \quad (*)$$

This pairwise route *fails* inside a slice. Witness: $n = 6$, $m = 2$, $\mu = (0, 3)$, $\lambda = (4, 5)$, slice $S_\nu = 5$, $\nu = (0, 5)$ and $\rho = (2, 3)$; then $f_\nu = \mathbf{1}_{[0,4]}$ and $f_\rho = \delta_2$, and at $s = 1$ the left side of $(*)$ is 0 while the right side is 1. Over $n \leq 6$, $m \geq 2$, $d \leq 7$, 4 of 9871 pairs fail, and 244 of 351 distinct pairs are not TP_2 -comparable in either orientation. *Scope*: the slice family is a genuine antichain; the failing pair is repaired only by the intervening element $\nu = (1, 4)$ of the same slice. Any proof must see the whole slice at once.

2. **The statement is not a general fact about difference-constraint polytopes.** The set in (P1) is cut out by constraints $x_p - x_q \leq c$ alone. But the corresponding general assertion — for $P = \{x \in \mathbb{Z}^{2m} : x_p - x_q \leq c_{pq}, \text{ box}\}$, the fibre count $v \mapsto \#\{x \in P : S_\kappa = u, S_\sigma = v\}$ is log-concave — is *false*: 312 failures out of 6474 fibres in a random sweep with $m = 2$; e.g. box $0 \leq x \leq (1, 1, 2, 4)$ with constraints $x_3 - x_1 \leq 1$, $x_3 - x_1 \leq 0$, $x_1 - x_0 \leq 3$, $x_1 - x_0 \leq 2$, $x_2 - x_3 \leq 1$ gives $(2, 3, 1, 1)$ at $u = 1$. *Scope*: the cylindric interlacing pattern, not mere difference-constraint structure, is load-bearing.
3. f_b **is not real-rooted**, so Newton's inequalities are unavailable: already $f_b = 1 + x + x^2$ occurs ($n = 3$, $\mu = (0)$, $\lambda = (2)$, $b = 0$). 600 of 1258 slices are not real-rooted.
4. f_b **is not a convolution of interval indicators.** 91 of 1176 slice polynomials are not products of cyclotomics; the smallest is $2 + 3x + 2x^2$ ($n = 5$, $\mu = (0, 2)$, $\lambda = (2, 4)$), which is irreducible over $\mathbb{Q}$.
5. **Concavity is false** — see Section 4. This matters because the proof of Proposition 3.5 below proves concavity, so it *cannot* extend to the general case.

3 The case $m = 2$

From here on $m = 2$, $\mu = (\mu_1, \mu_2)$, $\lambda = (\lambda_1, \lambda_2)$, and we abbreviate

$$A = \lambda_2 - n + 1, \quad B = \lambda_1, \quad C = \lambda_1 + 1, \quad D = \lambda_2,$$

so that $C = B + 1$ and $A = D - n + 1$. Fix b and put $u = |\mu| + b$. The slice $\{\nu \in \text{Box}(\mu) : S_\nu = u\}$ is parametrised by $t = \nu_1$, $\nu_2 = u - t$, with $t \in T = [T_-, T_+]$, $T_- = \max(\mu_1, u - \mu_1 - n + 1)$, $T_+ = \min(\mu_2 - 1, u - \mu_2)$. Given t , Proposition 2.1 says κ_1 and κ_2 range independently over

$$\kappa_1 \in [a_t, b_t] := [\max(t, A), \min(u - 1 - t, B)], \quad \kappa_2 \in [c_t, d_t] := [\max(u - t, C), \min(t + n - 1, D)].$$

Writing $s = \kappa_1 + \kappa_2 = u + a$, we must show that

$$G(s) = \sum_{t \in T} (\mathbf{1}_{[a_t, b_t]} * \mathbf{1}_{[c_t, d_t]})(s) \tag{1}$$

is PF_2 .

3.1 Exact reduction to a two-profile lemma

Proposition 3.1. Define $\Phi : [A, B] \rightarrow \mathbb{Z}$ and $\Psi : [C, D] \rightarrow \mathbb{Z}$ by

$$\Phi(i) = \min(i, u - 1 - i, T_+), \quad \Psi(j) = \max(u - j, j - n + 1, T_-).$$

Then Φ is concave and Ψ is convex, both are 1-Lipschitz, and

$$G(s) = \sum_{i+j=s} (\Phi(i) - \Psi(j) + 1)_+, \quad (2)$$

the sum being over $i \in [A, B]$, $j \in [C, D]$.

Proof. $\kappa_1 = i$ is admissible for the parameter t iff $A \leq i \leq B$, $t \leq i$ and $t \leq u - 1 - i$; together with $t \in T$ this says $T_- \leq t \leq \Phi(i)$. Likewise $\kappa_2 = j$ is admissible iff $C \leq j \leq D$ and $t \geq u - j$, $t \geq j - n + 1$, i.e. $\Psi(j) \leq t \leq T_+$. Hence the number of $t \in T$ for which the pair (i, j) is counted is $\#\{t : \Psi(j) \leq t \leq \Phi(i)\} = (\Phi(i) - \Psi(j) + 1)_+$, giving (2). Φ is a minimum of three affine functions of slope 1, -1 , 0 , hence concave and 1-Lipschitz; Ψ is a maximum of three affine functions of slope -1 , 1 , 0 , hence convex and 1-Lipschitz. $\square$

This is an *exact* reduction: nothing about cylindric shapes survives in (2) except the four numbers A, B, C, D and the two numbers u, n , all absorbed into Φ, Ψ . So condition (A) at $m = 2$ is implied by:

Conjecture 3.2 (Lemma T). *Let Φ be concave and 1-Lipschitz on an integer interval, Ψ convex and 1-Lipschitz on an integer interval (both extended by $\mp\infty$). Then $s \mapsto \sum_{i+j=s} (\Phi(i) - \Psi(j) + 1)_+$ is log-concave.*

Remark 3.3 (the 1-Lipschitz hypothesis is sharp). Both weakenings fail.

- Without any slope bound: $\Phi = (-1, 0, 1)$, $\Psi = (-3, 0, 4)$ on $\{0, 1, 2\}$ give $G = (3, 4, 6, 2)$, and $4^2 = 16 < 3 \cdot 6 = 18$. (40 failures in 19133 random trials.)
- With slopes in $\{-2, \dots, 2\}$: $\Phi = (3, 5, 5)$, $\Psi = (3, 1, 2, 3, 4)$ give $G = (1, 6, 10, 10, 7, 5, 2)$, which fails at $s = 4$: $7^2 = 49 < 10 \cdot 5 = 50$. (117 failures in 32900 random trials.)
- With slopes in $\{-1, 0, 1\}$: 0 failures in 29251 random trials, and 0 failures in an *exhaustive* enumeration of all such profiles of length ≤ 6 with base shifts in $[-4, 6]$ (29000 instances).

So the negative control bites, and the hypothesis is load-bearing rather than decorative.

3.2 The two breakpoints and the four regions

The four endpoint functions in (1) are each a min or max of two affine functions, and — this is the structural point — they share only *two* breakpoints:

$$\tau_1 := A = D - n + 1 \quad (\text{for } a_t \text{ and } d_t), \quad \tau_2 := u - 1 - B = u - C \quad (\text{for } b_t \text{ and } c_t).$$

Indeed $a_t = \max(t, A)$ switches at $t = A$ and $d_t = \min(t + n - 1, D)$ switches at $t = D - n + 1 = A$; while $b_t = \min(u - 1 - t, B)$ switches at $t = u - 1 - B$ and $c_t = \max(u - t, C)$ switches at $t = u - C = u - 1 - B$. Accordingly T meets at most four cells:

region	condition on t	a_t	b_t	c_t	d_t
I	$t < \tau_1, t \leq \tau_2$	A	B	$u - t$	$t + n - 1$
II	$\tau_1 \leq t \leq \tau_2$	t	B	$u - t$	D
III	$t \geq \tau_1, t > \tau_2$	t	$u - 1 - t$	C	D
IV	$t < \tau_1, t > \tau_2$	A	$u - 1 - t$	C	$t + n - 1$

Write $G = G_I + G_{II} + G_{III} + G_{IV}$ for the corresponding partial sums in (1).

Proposition 3.4 (regions I and III: a fixed factor). *Let $T_I = [p_0, p_1]$ be the (possibly empty) set of $t \in T$ in region I. Then*

$$G_I = \mathbf{1}_{[A, B]} * w_I, \quad w_I(x) = (p_1 - \max(u - x, x - n + 1, p_0) + 1)_+,$$

and $G_I \in \text{PF}_2$. Symmetrically, with $T_{III} = [q_0, q_1]$,

$$G_{III} = w_{III} * \mathbf{1}_{[C, D]}, \quad w_{III}(x) = (\min(x, u - 1 - x, q_1) - q_0 + 1)_+,$$

and $G_{III} \in \text{PF}_2$.

Proof. In region I the first interval $[a_t, b_t] = [A, B]$ does not depend on t , so

$$G_I = \sum_{t \in T_I} \mathbf{1}_{[A, B]} * \mathbf{1}_{[u-t, t+n-1]} = \mathbf{1}_{[A, B]} * \left(\sum_{t \in T_I} \mathbf{1}_{[u-t, t+n-1]} \right).$$

Now $u - t \leq x \leq t + n - 1$ is equivalent to $t \geq \max(u - x, x - n + 1)$, so the inner sum is $w_I(x) = \#\{t \in [p_0, p_1] : t \geq \max(u - x, x - n + 1)\}$, which is the displayed expression. The function $x \mapsto \max(u - x, x - n + 1, p_0)$ is convex, hence $x \mapsto p_1 - \max(\dots) + 1$ is concave, and w_I is its positive part, so $w_I \in \text{PF}_2$ by Lemma 1.3. An interval indicator is PF_2 , so $G_I \in \text{PF}_2$ by (P3).

In region III the second interval $[c_t, d_t] = [C, D]$ is constant and $t \leq x \leq u - 1 - t$ is equivalent to $t \leq \min(x, u - 1 - x)$; the same argument applies, now with $x \mapsto \min(x, u - 1 - x, q_1)$ concave. $\square$

Proposition 3.5 (regions II and IV: a constant support). *In region II one has $a_t + c_t = u$ and $b_t + d_t = B + D$ for every t ; in region IV one has $a_t + c_t = A + C$ and $b_t + d_t = u + n - 2$ for every t . Consequently, writing $(\ell, h) = (u, B + D)$ for region II and $(\ell, h) = (A + C, u + n - 2)$ for region IV, and $H(s) = \min(s - \ell + 1, h - s + 1)$,*

$$G_R(s) = \rho_R(H(s)), \quad \rho_R(v) = \sum_{t \in T_R} \min(v, n_t, m_t)_+,$$

where $n_t = b_t - a_t + 1$, $m_t = d_t - c_t + 1$. Both G_{II} and G_{IV} are concave on their support, hence PF_2 .

Proof. Region II: $a_t + c_t = t + (u - t) = u$ and $b_t + d_t = B + D$. Region IV: $a_t + c_t = A + C$ and $b_t + d_t = (u - 1 - t) + (t + n - 1) = u + n - 2$. In either case all the trapezoids $\mathbf{1}_{[a_t, b_t]} * \mathbf{1}_{[c_t, d_t]}$ have the same support $[\ell, h]$, and the convolution of two interval indicators of widths n_t, m_t is

$$(\mathbf{1}_{[a_t, b_t]} * \mathbf{1}_{[c_t, d_t]})(s) = \min(s - \ell + 1, h - s + 1, n_t, m_t)_+ = \min(H(s), n_t, m_t)_+.$$

Summing over $t \in T_R$ gives $G_R(s) = \rho_R(H(s))$, using $\min(\alpha, \beta)_+ = \min(\alpha_+, \beta_+)$.

Now $v \mapsto \min(v, n_t, m_t)_+$ is nondecreasing and concave on $v \geq 0$, so ρ_R is nondecreasing and concave on $[0, \infty)$ with $\rho_R(0) = 0$. The function H is concave, and $H(s) \geq 1$ for $s \in [\ell, h]$. A nondecreasing concave function composed with a concave function is concave, so $G_R = \rho_R \circ H$ is concave on $[\ell, h]$; it is nonnegative there and its support is an interval. Apply Lemma 1.2. $\square$

Theorem 3.6. *At $m = 2$, for every cylindric skew shape, every level n and every b , the sequence G of (1) decomposes as $G = G_I + G_{II} + G_{III} + G_{IV}$ with each nonzero summand in PF_2 , by Propositions 3.4 and 3.5.*

Corollary 3.7 (single-region criterion). *Let $T = [T_-, T_+]$ be the set of t for which both intervals in (1) are nonempty. If*

$$(T_+ < \tau_1 \text{ and } T_+ \leq \tau_2) \text{ or } (T_- \geq \tau_1 \text{ and } T_+ \leq \tau_2) \text{ or } (T_- \geq \tau_1 \text{ and } T_- > \tau_2) \text{ or } (T_+ < \tau_1 \text{ and } T_- > \tau_2)$$

— that is, if T lies in a single one of the four cells — then $k(\cdot, b)$ is PF_2 ; in particular condition (A) holds for this $(\lambda/\mu, n, b)$.

In a sweep over $2 \leq n \leq 9$, $m = 2$, $d \leq 12$ (5187 nonempty slices) the criterion of Corollary 3.7 is satisfied by 4147 slices, i.e. 79.9%.

3.3 The gap

What remains is exactly the mixed terms. By the symmetrisation identity recalled in (*), Theorem 3.6 yields log-concavity of G as soon as (*) holds for every pair of *distinct* regions $R \neq R' \in \{I, II, III, IV\}$:

$$2G_R(s)G_{R'}(s) \geq G_R(s-1)G_{R'}(s+1) + G_R(s+1)G_{R'}(s-1) \quad \text{for all } s. \quad (3)$$

Status of (3): verified, not proved. Over $2 \leq n \leq 8$, $m = 2$, $d \leq 10$ (2655 slices), (3) holds on all 525 occurring region pairs: I–II 177/177, II–III 177/177, I–III 81/81, I–IV 45/45, III–IV 45/45.

Two structural facts constrain any proof of (3). First, all four supports are nested inside the common maximal support: at $t = \tau_1$ the region I support $[A + u - t, B + n - 1 + t]$ equals $[u, B + D]$, which is exactly the region II support, and at $t = \tau_2$ the region III support $[t + C, u - 1 - t + D]$ also equals $[u, B + D]$. Second, (3) is a statement about a *cancellation across the regions*: it is *not* implied by any slice-local comparison, since (*) already fails between individual slice members (Section 2.1, item 1).

4 Concavity is false

Proposition 3.5 proves more than log-concavity for regions II and IV: it proves *concavity*. This cannot be the general mechanism.

Proposition 4.1. *The sequence $a \mapsto k(a, b)$ need not be concave on its support, already at $m = 2$.*

Proof. At $n = 8$, $m = 2$, $\mu = (0, 3)$, $\lambda = (4, 7)$, $b = 2$ one finds $k(\cdot, 2) = (1, 3, 6, 7, 6, 3, 1)$; concavity fails at the second entry, where $2 \cdot 3 = 6 < 1 + 6 = 7$. (Log-concavity holds there: $3^2 = 9 \geq 1 \cdot 6 = 6$.) At $n = 7$, $m = 3$, $\mu = (0, 1, 4)$, $\lambda = (2, 3, 6)$, $b = 2$ one finds $k(\cdot, 2) = (1, 3, 6, 3, 1)$, and again $2 \cdot 3 = 6 < 1 + 6 = 7$. $\square$

Over a sweep with $2 \leq n \leq 9$, $m \leq 7$, $d \leq 11$ (96781 interior points), concavity fails at 1758 points — distributed over $m = 2$ (44), $m = 3$ (964), $m = 4$ (720), $m = 5$ (30) — while log-concavity fails at 0. So concavity is a small-case artifact and log-concavity is the correct statement; in particular the concave-composition argument of Proposition 3.5 *must* break when more than one region is active, which is consistent with the gap being exactly (3).

5 What $m \geq 3$ needs

Proposition 2.1 is valid for every m . The special feature of $m = 2$ is that the box-slice $\{\nu \in \text{Box}(\mu) : S_\nu = u\}$ is one-dimensional, so the whole family is indexed by a single parameter t and the four endpoint functions have only two breakpoints. For $m \geq 3$ the slice has dimension $m - 1$ and no such parametrisation exists; neither the reduction of Proposition 3.1 nor the region decomposition of Theorem 3.6 is available. We record this as the precise scope of the present work rather than as a technical remark: the results of Section 3 are statements about $m = 2$ and nothing in their proofs suggests an m -independent form.

6 Brändén–Huh: what the closure theorems do and do not give

The programme had recorded “read Brändén–Huh 1902.03719 at source” as its highest-value next step. This was done. The findings relevant here, cited by internal L^AT_EX label (an e-print theorem *number* is a fact about a build we do not control):

- Thm \label{flow}: if $f \in L_n^d$ then $f(Av) \in L_m^d$ for any nonnegative $n \times m$ matrix A — no positivity, invertibility or squareness.
- Cor \label{derivatives}: $\sum_i a_i \partial_i f \in L_n^{d-1}$ for $a_i \geq 0$.
- Cor \label{CorollaryProduct}: **Lorentzian polynomials are closed under products**, and Cor \label{CorollaryConvolution}: $N(f), N(g)$ Lorentzian $\Rightarrow N(fg)$ Lorentzian. This was *not* known to the programme and the brief explicitly disclaimed it.
- Closure under *addition* is never addressed in the paper; it is false, and the registry already holds the witness $((x + y)^2 + (2x + y)^2)$.
- There is *no* characterisation of Lorentzian polynomials as limits of volume polynomials; that was Question \label{RealizationQuestion}, answered negatively in an added-in-proof footnote. So the “realise $N(s^c)$ as a volume polynomial” route has no theorem behind it.
- Cor \label{normalizedcoefficients}: if $\nu_f(\alpha) = \log c_\alpha$ is an M-concave function then $f = \sum (c_\alpha / \alpha!) w^\alpha$ is Lorentzian. Sufficient, not necessary.
- The paper proves nothing whatever about Schur polynomials.

Two consequences for this programme. *Negative*: none of these shortcuts condition (A). One might hope to obtain $h_b(x, z) = [y^b] s^c(x, y, z)$ by applying ∂_y^b (Cor **derivatives**) and then $y = 0$ (the $\theta = 0$ case of Thm **flow**) to $N(s^c)$ — but that presupposes $N(s^c)$ Lorentzian, which is the whole conjecture. Condition (A) is a strictly weaker necessary condition and must be proved on its own. *Positive*: Cor **normalizedcoefficients** gives a genuinely new route to the *main* conjecture, since (L2) — M-convexity of $\text{supp } K^c$ — is already proved, so the local form of M-concavity (only $|\alpha - \beta|_1 = 4$ needs checking) applies. At $\ell = 3$ the case $\alpha - \beta = (2, -2, 0)$ of that local axiom *is* exactly condition (A). This is recorded as a new node, not pursued here.

7 Verification

All arithmetic is exact integer arithmetic. Two independent implementations of K^c at $\ell = 3$ were written for this session — the box model of Proposition 2.1 and a direct enumeration of chains $\mu \prec \kappa \prec \sigma \prec \lambda$ — and agree on 647/647 instances ($2 \leq n \leq 6$, $1 \leq m \leq n$, $d \leq 6$).

claim	scope	result
k via box model = k via chains	$n \leq 6, m \leq n, d \leq 6$	647/647
Proposition 2.1 reproduces k	$n \leq 5, m \leq n, d \leq 6$	227/227
condition (A)	$n \leq 9, m \leq 7, d \leq 11$	96781/96781 interior points
Proposition 3.1 exact	$n \leq 9, m = 2, d \leq 11$	6464 slices, exact
Φ concave, Ψ convex, both 1-Lipschitz	same	0 violations
Conjecture 3.2, exhaustive	profiles len ≤ 6 , shifts $[-4, 6]$	29000/29000
Conjecture 3.2, random	slopes $\{-1, 0, 1\}$	29251/29251
four-region closed forms	$n \leq 9, m = 2, d \leq 12$	I 1363/1363, II 2988/2988, III 1363/1363, IV 579/579
$G = \sum_R G_R$	$n \leq 8, m = 2, d \leq 10$	2655/2655
each nonempty region PF_2	same	0 failures
single-region criterion correct	$n \leq 9, m = 2, d \leq 12$	5187/5187; 4147 single
inter-region (3)	same	525/525 (gap)
NEGATIVE CONTROLS — each bites, so the hypotheses are load-bearing		
Conjecture 3.2 with slopes ± 2	random	117 failures / 32900
Conjecture 3.2, no slope bound	random	40 failures / 19133
pairwise (*) inside a slice	$n \leq 6, m \geq 2, d \leq 7$	4 failures / 9871
general difference-constraint polytope	random, $m = 2$	312 failures / 6474
concavity of $k(\cdot, b)$	$n \leq 9, m \leq 7, d \leq 11$	1758 failures / 96781
f_b real-rooted	$n \leq 6, m \leq n, d \leq 7$	600 failures / 1258
f_b product of cyclotomics	same	91 failures / 1176
greedy induction on t	random	fails 47%
greedy induction on columns	random	fails 69%

8 Summary

- **Proved, all m :** condition (A) is equivalent to log-concavity of a sum of PF_2 sequences over one slice of a box (Proposition 2.1).
- **Proved, $m = 2$:** an exact reduction to two 1-Lipschitz profiles (Proposition 3.1); a four-region decomposition with each region PF_2 (Theorem 3.6); hence condition (A) whenever a single region is active (Corollary 3.7), 79.9% of slices in our sweep.
- **Gap, precisely:** inequality (3) between distinct regions. Verified 525/525; unproved.
- **Open:** $m \geq 3$ entirely, beyond Proposition 2.1.
- **Not claimed:** (A) does not settle (L3). By 13-det-reduction, (L3) at $\ell = 3$ needs $e_2(M) \leq 0$ and $\det M \geq 0$; (A) gives only the former, and the latter is condition (B).